

Lab 3: Component Modelling & Architectural Pattern Selection

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System: EV Charging Reservation System

Architecture Selection

We chose Microservices Architecture for the EV Charging Reservation System. The system has different functionalities such as managing reservations, checking charger availability, processing payments and storing reservation details. These functionalities are separated into individual components which communicate through well-defined interfaces.

Reasons for Choosing Microservices

1. Independent scaling:

The number of requests received by each service may not always be the same. For example, a large number of users may check real-time charger availability even if they do not make a reservation. Using microservices allows the Charger Service Availability component to be scaled independently without having to scale the complete application.

2. Fault isolation and easier maintenance:

Since the major functionalities are separated into different services, failure of one service does not necessarily bring down the complete system. For example, if the Payment Service is temporarily unavailable, charger availability can still be viewed. Individual services can also be modified or deployed without making major changes to the other components.

Security Advantage

Payment and reservation information contains sensitive user data. With a microservices architecture, the Payment Service and Reservation Database can be separated from the user-facing components. Communication between these services can be restricted using authenticated and authorized API requests, which reduces unnecessary access to sensitive data.

Performance Benefit

Services with higher workloads can be scaled independently according to demand. In this system, real-time charger availability may receive frequent requests from multiple EV drivers. Scaling the Charger Service Availability separately helps maintain faster response times without increasing resources for all the other services.

Conclusion

Microservices Architecture is suitable for the EV Charging Reservation System because it provides independent scaling, fault isolation, better separation of responsibilities and controlled access between services. These features support the real-time availability, reservation and payment requirements of the system.